

Gadha Krishna Leons

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PROFESSIONAL SUMMARY

Final-year Ph.D. researcher in Cancer Genomics specializing in the molecular characterization of B-cell Acute Lymphoblastic Leukemia (B-ALL). Proven track record of integrating high-throughput transcriptomics (RNA-Seq) with clinical metadata to identify cryptic subtypes and predictive biomarkers in BCR-ABL1-negative cohorts. Expertise in developing custom bioinformatics pipelines (R, Python, WGCNA) and validating in-silico findings through rigorous wet-lab assays (Sanger, MLPA, qPCR). Aiming to translate large-scale genomic data into actionable biological insights by combining bioinformatics pipelines with molecular biology in a postdoctoral role.

EDUCATION

- Ph.D. in Cancer Genomics** 2022 – 2026
All India Institute of Medical Sciences (AIIMS) New Delhi, India
 - Study of predictive and prognostic molecular genetic aberrations in BCR::ABL1 negative B-ALL.
- Advanced PG Diploma in Life Science Technologies** 2020 – 2021
SRM-DBT Platform, SRMIST Chennai, India
- M.Sc. in Applied Genetics** 2017 – 2019
Bangalore University Bangalore, India
 - CGPA: 7.8 (1st Class Distinction)
- B.Sc. (Genetics, Microbiology, Biochemistry)** 2014 – 2017
Ramaiah College of Arts, Science and Commerce Bangalore, India
 - CGPA: 8.42 (1st Class Exemplary)

RESEARCH EXPERIENCE

- Junior Research Fellow / Ph.D. Scholar** Oct 2021 - Present
AIIMS, New Delhi New Delhi, India
 - Led comprehensive transcriptomic profiling of B-ALL samples to identify high-risk subtypes (Ph-like) and novel fusions.
 - Engineered and optimized bioinformatics pipelines using **R** and **Python** to analyze RNA-Seq data (**DGE analysis, WGCNA**).
 - Validated in-silico findings using wet-lab techniques including **Sanger sequencing, RT-qPCR**, and **MLPA**.
 - Discovered prognostic biomarkers by correlating transcriptomic signatures with clinical survival data.
- PG Diploma Researcher** Jan 2021 - July 2021
SRM-DBT Platform, SRMIST Chennai, India
 - Conducted phylogenetic analysis and clustering of SARS-CoV-2 genomes circulating in the South Indian population.
 - Performed molecular docking studies to model protein-protein interactions and analyze spike protein variants.
- Master's Dissertation Researcher** Jan 2019 - July 2019
Bangalore University Bangalore, India
 - Characterized sperm morphological diversity in Caecilians (Amphibia: Gymnophiona) of Western Ghats.

TECHNICAL SKILLS

- Bioinformatics

- **NGS Analysis:** Bulk RNA-Seq Analysis (STAR, DESeq2), Single-Cell RNA-Seq Analysis (Seurat)
- **Methods & Concepts:** Weighted Gene Coexpression Network Analysis (WGCNA), Pathway Enrichment (GSEA, ClusterProfiler, Enrichr, ssGSEA),
- **Structural:** Protein Structure Modeling, Protein-Protein Docking

- Programming

- R (Bioconductor, Tidyverse, ggplot2), Python, Linux/Unix, Git/Version Control

- Molecular Biology (Wet Lab)

- NGS Library Prep (Illumina), Sanger Sequencing, MLPA, RT-qPCR, Flow Cytometry, DNA/RNA Extraction
- **General Research:** Cell Culture, Zebrafish Genetic Engineering (CRISPR, Microinjection)

PUBLICATIONS

- [1] **Leons, G. K., Gupta, S. K., Sharma, P., et al. (In Review). Decoding the 'B-Other' Black Box: Integrated RNA-Sequencing and Machine Learning Unveils Cryptic B-ALL Subtypes in a Single Centre Study.**
- [2] Gupta, S. K., **Leons, G. K., Sharma, P., et al. (2025). Application of Targeted RNA-Sequencing in High-Risk B-Cell Acute Lymphoblastic Leukemia (B-ALL).** *Int J Lab Hematol.* DOI: 10.1111/ijlh.14551
- [3] Sharma, P., **Leons, G. K., Singh, V., et al. (2025). Gene expression signature of IKZF1 deleted B-cell acute lymphoblastic leukaemia reveals a unique regulation of MUC4.** *Br J Haematol.* DOI: 10.1111/bjh.70014
- [4] Gupta, S.K., **Leons, G. K. (2025). Genetics of B-Cell Acute Lymphoblastic Leukemia (B-ALL): Recent Updates and Indian Perspective.** *Indian J Hematol Blood Transfus.* DOI: 10.1007/s12288-025-01988-y
- [5] **Leons, G. K., Sharma, P., Gupta, S. K., et al. (2023). Detection of a novel SNX2 gene breakpoint in the SNX2::ABL1 fusion transcript in Ph-like B-cell acute lymphoblastic leukemia.** *Pediatric Blood & Cancer.* DOI: 10.1002/pbc.30546
- [6] Sharma, P., Gupta, S. K., **Leons, G. K., et al. (2022). IKZF1::CIITA, a novel fusion transcript in a case of relapsed B-cell acute lymphoblastic leukemia.** *Pediatric blood & cancer*, 70(1), e29872. DOI: 10.1002/pbc.29872
- [7] Gupta, S. K., & **Leons, G. K. (2025). Molecular Diagnostics.** In Sharma SK (Ed.), *Manual of Hematology* (pp. 158-169). Jaypee Brothers Medical Publishers.
- [8] Gupta, S. K., & **Leons, G. K. (2025). Diagnosis of Ph-like B-ALL: The Ideal Approach in the Indian Setting.** In *Recent Advances in Hematology-5* (pp. 92-103). Jaypee Brothers Medical Publishers.

CONFERENCE PRESENTATIONS

- Poster Presentation

2026
2026 Korean Society of Hematology (KSH) International Conference & 67th Annual Meeting Seoul, Korea
The Seeds of Relapse: Uncovering the Dormant Transcriptomic Signatures of Future Relapse at Diagnosis

- Oral Presentation

2025
87th Annual Meeting of the Japanese Society of Hematology (JSH) Kobe, Japan
Transcriptomic Signatures and Fusion Landscape in Relapsed B-ALL: Insights from RNA-Sequencing

- **Poster Presentation** 2025
Molecular Analysis for Precision Oncology Congress (ESMO-MAP) Paris, France
 Transcriptomic Profiling of CDKN2A/B-Deleted B-ALL Reveals Key Coding and Non-Coding Biomarkers
- **Oral Presentation** 2024
2024 Korean Society of Hematology (KSH) International Conference & 65th Annual Meeting Seoul, Korea
 Adverse Clinical Impact of RB1 Gene Deletions Alone and with Concurrent IKZF1 Gene Deletions in Pediatric B-ALL
- **Abstract Publication** 2025
European Hematology Association (EHA)
 Whole Transcriptome Sequencing Uncovers Key Coding and Non-Coding Biomarkers in IKZF1Plus B-ALL.

HONORS AND AWARDS

- **ICKSH Travel Award** 2026
2026 Korean Society of Hematology (KSH) International Conference & 67th Annual Meeting (ICKSH)
- **JSH Travel Award** 2025
87th Annual Meeting of the Japanese Society of Hematology (JSH)
- **ICMR International Travel Grant** 2025
Molecular Analysis for Precision Oncology (MAP) Congress-European Society for Medical Oncology (ESMO)
- **ANRF International Travel Grant** 2024
2024 Korean Society of Hematology (KSH) International Conference & 65th Annual Meeting (ICKSH)
- **JGEEBILS Qualified** 2021
Tata Institute of Fundamental Research (TIFR)
 ◦ Reference No.: GS2021BIOPHD030224
- **GATE Biotechnology (AIR 1289)** 2020
Indian Institute of Technology (IIT)
 ◦ Scored 475 in Graduate Aptitude Test in Engineering.

CERTIFICATIONS

- **NASA GeneLab | GL4U: RNA Sequencing Module Set** Aug 2025
- **NASA GeneLab | GL4U: Introduction Module Set** Jul 2025
- **Training in “Engineering Zebrafish with CRISPR Tools” (CSIR-CCMB)** Mar 2020
- **Certificate in Animal Cell Culture Techniques (CellKraft Biotech)** Feb 2020
- **Certificate in Genetic Counseling (IMAEVarsity& Martin Luther Christian University)**

REFERENCES

Dr. Sanjeev Kumar Gupta
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 AIIMS, New Delhi
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Dr. Ritu Gupta
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 AIIMS, New Delhi
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